

## Taiyang Xu

### Université catholique de Louvain

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|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Current position</b>           | <b>Université catholique de Louvain</b> , Louvain-la-Neuve, Belgium<br>Institut de Recherche en Mathématique et Physique<br>Postdoctoral Researcher (Mentor: Prof. Christophe Charlier)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 08/2026 – now                              |
| <b>Past positions</b>             | <b>Fudan University</b> , Shanghai, China<br>Department of Mathematics<br>Postdoctoral Fellow (Mentor: Prof. Lun Zhang)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 07/2024 – 06/2026                          |
| <b>Education</b>                  | <b>Fudan University</b> , Shanghai, China<br>Ph.D. in Mathematics, supervisor Prof. Engui Fan<br>Thesis title: “ <i>On the long-time asymptotics of the local and nonlocal mKdV equation under the nonzero background</i> ”<br><br><b>China University of Mining and Technology</b> , Xuzhou, China<br>B.Sc. in Mathematics, Distinguished Honor<br>Thesis title: “ <i>Inverse scattering theory and integrability on several kinds of nonlinear evolution equations</i> ”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 09/2019 – 06/2024<br><br>09/2015 – 06/2019 |
| <b>Research interests</b>         | Integrable ODEs/PDEs, Random matrix theory, Determinantal point processes, Orthogonal polynomials, Asymptotic analysis, Riemann-Hilbert problems, Special functions ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                            |
| <b>Publications and preprints</b> | Listed in reverse chronological order of first public availability.<br>10. Critical asymptotics of the Camassa-Holm equation in the small dispersion limit at the point of gradient catastrophe (with Yiling Yang and Lun Zhang)<br><i>available upon request</i><br>9. Moment generating function of the tacnode process<br><i>arXiv:2606.17771</i><br>8. Confluent hypergeometric kernel determinant on multiple large intervals (with Lun Zhang and Zhengyang Zhao)<br><i>arXiv:2508.10463</i><br>7. Transient asymptotics of the modified Camassa-Holm equation (with Yiling Yang and Lun Zhang)<br><i>Journal of the London Mathematical Society</i> , 110 (2024), e12967.<br>6. Painlevé transcendents in the defocusing mKdV equation with non-zero boundary conditions (with Engui Fan and Zhaoyu Wang)<br><i>Communications in Mathematical Physics</i> , 406 (2025), 181.<br>5. Large-time asymptotics to the focusing nonlocal modified Kortweg-de Vries equation with step-like boundary conditions (with Engui Fan)<br><i>Studies in Applied Mathematics</i> , 150 (2023), 1217–1273.<br>4. On the large-time asymptotics of the defocusing mKdV equation with step-like initial data (with Yidan Zhang)<br><i>Mathematical Physics, Analysis and Geometry</i> , 29 (2026), 38.<br>3. On the Cauchy problem of defocusing mKdV equation with finite density initial data: long-time asymptotics in soliton-less regions (with Engui Fan and Zechuan Zhang)<br><i>Journal of Differential Equations</i> , 372 (2023), 55–122.<br>2. Soliton resolution and asymptotic stability of $N$ -soliton solutions for the defocusing mKdV equation with a non-vanishing background (with Engui Fan and Zechuan Zhang)<br><i>Physica D: Nonlinear Phenomena</i> , 472 (2025), 134526. |                                            |

1. Riemann-Hilbert approach for multisoliton solutions of generalized coupled fourth-order nonlinear Schrödinger equations (with Weiqi Peng and Shoufu Tian)  
*Mathematical Methods in the Applied Sciences*, 43 (2020), 865–880.

## Grants

- Shanghai Post-doctoral Excellence Program (Grant No. 2024100) 07/2024 – 06/2026  
Project: Riemann-Hilbert method for several asymptotic problems related to universality from integrable systems and random matrix theory  
Role: Principal Investigator
- China Postdoctoral Science Foundation (Grant No. 2024M760480) 07/2024 – 06/2026  
Project: Semiclassical asymptotics and universality for nonlinear integrable shallow water wave systems  
Role: Principal Investigator

## Teaching

### 09/2019 – 07/2026 @Fudan

- Fall, 2025: Recitation & Tutorial Instructor of Mathematical Analysis III (MATH130001.04).
- Spring, 2024: TA of Methods of Asymptotic Analysis (MATH630117).
- Fall, 2021: TA of Calculus A (MATH120021.02).
- Spring, 2020: TA of Calculus B (MATH120004.01) (Online).
- Fall, 2019: TA of Calculus B (MATH120003.01).

## Research talks

- “*On the Fredholm determinants*” (invited), Seminar @Jiangsu Normal University, Xuzhou, China, 23 Jun, 2026.
- “*Dubrovin’s conjecture for the Camassa-Holm equation*” (invited), 3rd Workshop on Integrable Systems and Random Matrix Theory 2026, Dongguan, China, 4–17 Jan, 2026.
- “*Confluent hypergeometric kernel determinant on multiple large intervals*” (invited), Bielefeld-Melbourne-Seoul Online Random Matrix Seminar, 15th Oct, 2025.
- “*Confluent hypergeometric kernel determinant on multiple large intervals*” (contributed), 5th ZiF Summer School “Randomness in Physics and Mathematics: From Thermalisation in Quantum Systems to Random Matrices”, Universität Bielefeld, Bielefeld, Germany, 25 Aug–6 Sep, 2025.
- “*An introduction to Dubrovin’s universality conjecture for integrable PDEs, and some progress on the Camassa-Holm equation*” (invited), Seminar @UCLouvain, Louvain-la-Neuve, Belgium, 21 Aug, 2025.
- “*Transient asymptotics of the modified Camassa-Holm equation*” (invited), The 2nd Workshop on Integrable Systems and Random Matrix Theory, Dongguan, China, 5–17 Jan, 2025.
- “*Integrable PDEs with nonzero boundary conditions: large-time asymptotics*” (contributed), The 15th Hemudu Forum on Integrable Systems, Ningbo, China, 24–26 Nov, 2023.

## Other presentations Outreach talks

- “*Some asymptotic problems in mathematical physics*” (invited), Shanghai Institute of Technical Physics, Shanghai, China, 29th Apr, 2025.

12 **presentations** (including reading reports) given at Fudan Integrable Systems and Random Matrix Theory Seminar.

## Academic visits

- 22/06/2026 – 25/06/2026, Jiangsu Normal University, China. (Host: Dong Yao)
- 18/08/2025 – 24/08/2025, UCLouvain, Belgium. (Host: Christophe Charlier)
- 31/03/2025 – 11/04/2025, Chongqing University, China. (Host: Yiling Yang)

(I greatly appreciate for their warm hospitality)

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| <b>Service</b>                 | <p><b>Co-organizer of the following activities</b></p> <ul style="list-style-type: none"> <li>– (with Lun Zhang) Weekly RMT seminar @Fudan, 09/2022 – 06/2026.</li> <li>– (with Junwen Liu and Lun Zhang) Mini-workshop on Random Matrices, Fudan University, Shanghai, China, 4 June, 2026.</li> <li>– (with Engui Fan, Zhaoyu Wang, Yiling Yang and Lun Zhang) Workshop on Integrable Systems, Fudan University, Shanghai, China, 28 Nov, 2025.</li> <li>– (with Lun Zhang) Mini-workshop on Asymptotic Analysis, Fudan University, Shanghai, China, 5–9 Jun, 2025.</li> </ul> <p><b>Referee services provided for the following journals</b></p> <ul style="list-style-type: none"> <li>– Analysis and Applications</li> <li>– Journal of Nonlinear Mathematical Physics</li> <li>– Qualitative Theory of Dynamical Systems</li> <li>– Studies in Applied Mathematics</li> </ul> <p><b>Other service</b></p> <ul style="list-style-type: none"> <li>– zbMATH Open Reviewer</li> </ul> |
| <b>Scholarships and awards</b> | <p><b>09/2019 – 06/2024 @Fudan (Doctorate)</b></p> <ul style="list-style-type: none"> <li>– Shanghai Outstanding Graduate (Graduation with Honors), 2024.</li> <li>– Huatai Securities Science and Technology Scholarship, 2023.</li> <li>– Pacific Insurance Company Scholarship, 2022.</li> <li>– Outstanding Doctoral Candidate Scholarship, Fudan University, 2021.</li> <li>– Fudan University Doctoral Scholarship, 2019–2023.</li> </ul> <p><b>09/2015 – 06/2019 @CUMT (Undergraduate)</b></p> <ul style="list-style-type: none"> <li>– Outstanding Undergraduates in China University of Mining and Technology, 2019.</li> </ul>                                                                                                                                                                                                                                                                                                                                                 |
| <b>Status</b>                  | <b>China</b> – citizen, <b>Married</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Languages</b>               | <ul style="list-style-type: none"> <li>– Chinese (native), The Qianwei dialect of Sichuanese (native)</li> <li>– English</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Computer skills</b>         | $\LaTeX$ , Mathematica, Matlab, HTML, C++, Javascript                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Wednesday 30<sup>th</sup> September, 2026